Отчет по практической работе №2

Студент: [Elyakova Alesya Lvovna]

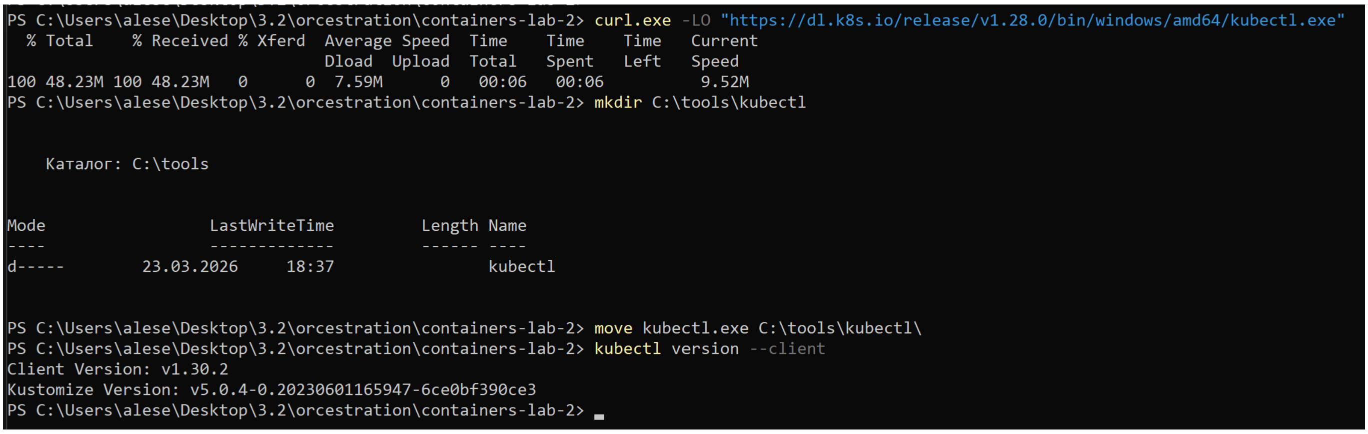
Группа: [16]

Дата выполнения: [20.04.2026]

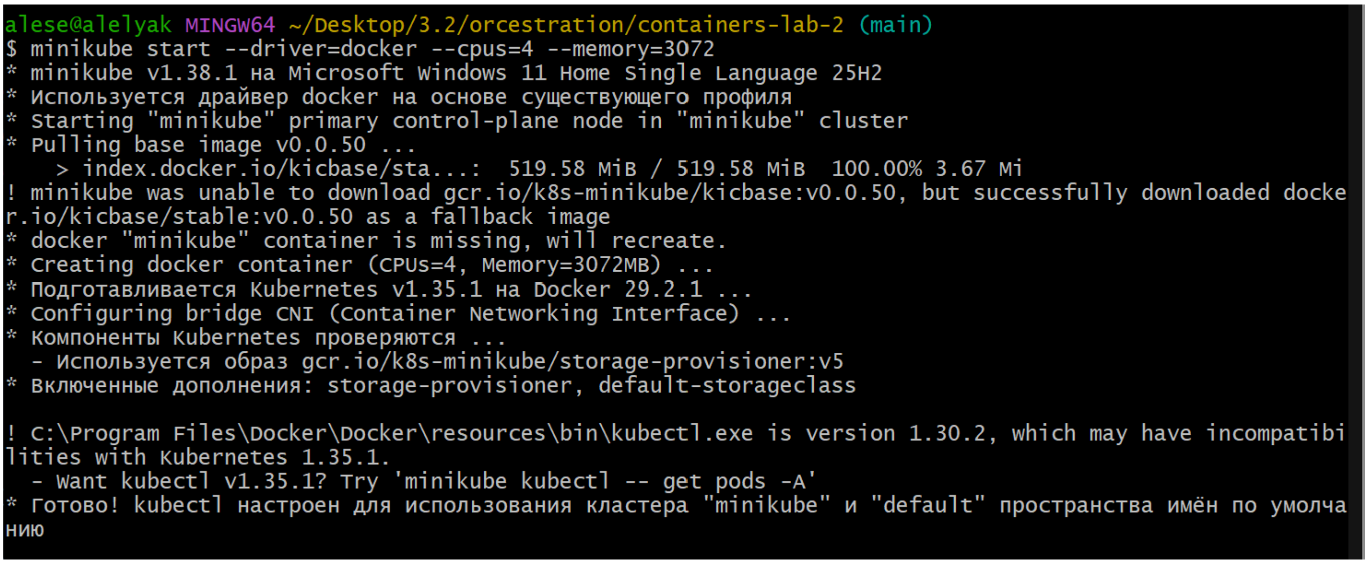
1. Информация о кластере

1.1 Статус Minikube

Проверка установки kubernetes



Запуск minikube



Статус minikube

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-2 (main)
$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Включение дополнений

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-2 (main)
$ minikube addons enable dashboard
* dashboard is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWN
ERS
  - Используется образ docker.io/kubernetesui/metrics-scraper:v1.0.8
  - Используется образ docker.io/kubernetesui/dashboard:v2.7.0
* Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

* The 'dashboard' addon is enabled

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-2 (main)
$ minikube addons enable metrics-server
* metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWN
ERS
  - Используется образ registry.k8s.io/metrics-server/metrics-server:v0.8.1
* The 'metrics-server' addon is enabled
```

Проверка создания секрета

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-2 (main)
$ kubectl get secrets
NAME          TYPE                             DATA   AGE
ghcr-secret   kubernetes.io/dockerconfigjson   1      5s
```

1.2 Узлы кластера

Информация о кластере

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:61537
CoreDNS is running at https://127.0.0.1:61537/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

Список узлов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl get nodes
NAME       STATUS   ROLES           AGE    VERSION
minikube   Ready    control-plane   103m   v1.35.1

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl get nodes -o wide
NAME       STATUS   ROLES           AGE    VERSION   INTERNAL-IP    EXTERNAL-IP   OS-IMAGE
      KERNEL-VERSION                     CONTAINER-RUNTIME
minikube   Ready    control-plane   103m   v1.35.1   192.168.49.2   <none>        Debian GNU/Linux 12 (b
ookworm)   6.6.87.2-microsoft-standard-WSL2   docker://29.2.1
```

Вывод kubectl describe nodes

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl describe nodes
Name:               minikube
Roles:              control-plane
Labels:             beta.kubernetes.io/arch=amd64
                    beta.kubernetes.io/os=linux
                    kubernetes.io/arch=amd64
                    kubernetes.io/hostname=minikube
                    kubernetes.io/os=linux
                    minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
                    minikube.k8s.io/name=minikube
                    minikube.k8s.io/primary=true
                    minikube.k8s.io/updated_at=2026_04_20T15_56_24_0700
                    minikube.k8s.io/version=v1.38.1
                    node-role.kubernetes.io/control-plane=
                    node.kubernetes.io/exclude-from-external-load-balancers=
Annotations:        node.alpha.kubernetes.io/ttl: 0
                    volumes.kubernetes.io/controller-managed-attach-detach: true
CreationTimestamp:  Mon, 20 Apr 2026 15:56:20 +0300
Taints:             <none>
Unschedulable:      false
Lease:
  HolderIdentity:  minikube
  AcquireTime:     <unset>
  RenewTime:       Mon, 20 Apr 2026 17:41:46 +0300
Conditions:
  Type             Status  LastHeartbeatTime                 LastTransitionTime                Reason                       M
essage
  ----             ------  -----------------                 ------------------                ------                       -
------
  MemoryPressure   False   Mon, 20 Apr 2026 17:39:14 +0300   Mon, 20 Apr 2026 15:56:16 +0300   KubeletHasSufficientMemory   k
ubelet has sufficient memory available
  DiskPressure     False   Mon, 20 Apr 2026 17:39:14 +0300   Mon, 20 Apr 2026 15:56:16 +0300   KubeletHasNoDiskPressure     k
ubelet has no disk pressure
  PIDPressure      False   Mon, 20 Apr 2026 17:39:14 +0300   Mon, 20 Apr 2026 15:56:16 +0300   KubeletHasSufficientPID      k
ubelet has sufficient PID available
  Ready            True    Mon, 20 Apr 2026 17:39:14 +0300   Mon, 20 Apr 2026 15:56:27 +0300   KubeletReady                 k
ubelet is posting ready status
```

```
Addresses:
  InternalIP:  192.168.49.2
  Hostname:    minikube
Capacity:
  cpu:                16
  ephemeral-storage:  1055762868Ki
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             7973056Ki
  pods:               110
Allocatable:
  cpu:                16
  ephemeral-storage:  1055762868Ki
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             7973056Ki
  pods:               110
System Info:
  Machine ID:                 86e0bfeb3f77427722393c2969964edb
  System UUID:                86e0bfeb3f77427722393c2969964edb
  Boot ID:                    5cba7501-2cf4-443a-82af-9adf00947fc9
  Kernel Version:             6.6.87.2-microsoft-standard-WSL2
  OS Image:                   Debian GNU/Linux 12 (bookworm)
  Operating System:           linux
  Architecture:               amd64
  Container Runtime Version:  docker://29.2.1
  Kubelet Version:            v1.35.1
  Kube-Proxy Version:
PodCIDR:                      10.244.0.0/24
PodCIDRs:                     10.244.0.0/24
Non-terminated Pods:          (11 in total)
```

```
  Namespace                   Name                                          CPU Requests  CPU Limits  Memory Requests  Memory
 Limits  Age
  ---------                   ----                                          ------------  ----------  ---------------  ------
-------  ---
  kube-system                 coredns-7d764666f9-4sjsh                      100m (0%)     0 (0%)      70Mi (0%)        170Mi
(2%)      105m
  kube-system                 coredns-7d764666f9-lnz4p                      100m (0%)     0 (0%)      70Mi (0%)        170Mi
(2%)      105m
  kube-system                 etcd-minikube                                 100m (0%)     0 (0%)      100Mi (1%)       0 (0%)
          105m
  kube-system                 kube-apiserver-minikube                       250m (1%)     0 (0%)      0 (0%)           0 (0%)
          105m
  kube-system                 kube-controller-manager-minikube              200m (1%)     0 (0%)      0 (0%)           0 (0%)
          105m
  kube-system                 kube-proxy-m6fmv                              0 (0%)        0 (0%)      0 (0%)           0 (0%)
          105m
  kube-system                 kube-scheduler-minikube                       100m (0%)     0 (0%)      0 (0%)           0 (0%)
          105m
  kube-system                 metrics-server-9d74bb658-mrbxp                100m (0%)     0 (0%)      200Mi (2%)       0 (0%)
          20m
  kube-system                 storage-provisioner                           0 (0%)        0 (0%)      0 (0%)           0 (0%)
          105m
  kubernetes-dashboard        dashboard-metrics-scraper-5565989548-zx7dh    0 (0%)        0 (0%)      0 (0%)           0 (0%)
          21m
  kubernetes-dashboard        kubernetes-dashboard-b84665fb8-nb7d9          0 (0%)        0 (0%)      0 (0%)           0 (0%)
          21m
Allocated resources:
  (Total limits may be over 100 percent, i.e., overcommitted.)
  Resource           Requests    Limits
  --------           --------    ------
  cpu                950m (5%)   0 (0%)
  memory             440Mi (5%)  340Mi (4%)
  ephemeral-storage  0 (0%)      0 (0%)
  hugepages-1Gi      0 (0%)      0 (0%)
  hugepages-2Mi      0 (0%)      0 (0%)
Events:              <none>
```

Список доступных API-ресурсов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl api-resources | findstr pod
pods                                po           v1                                true         Pod
podtemplates                                     v1                                true         PodTemplate
horizontalpodautoscalers            hpa          autoscaling/v2                    true         HorizontalPodAutoscaler
poddisruptionbudgets                pdb          policy/v1                         true         PodDisruptionBudget

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl api-resources | findstr deployment
deployments                         deploy       apps/v1                           true         Deployment
```

Просмотр конфигурации

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl config view
apiVersion: v1
clusters:
- cluster:
    certificate-authority: C:\Users\alese\.minikube\ca.crt
    extensions:
    - extension:
        last-update: Mon, 20 Apr 2026 15:56:24 MSK
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: cluster_info
    server: https://127.0.0.1:61537
  name: minikube
contexts:
- context:
    cluster: minikube
    extensions:
    - extension:
        last-update: Mon, 20 Apr 2026 15:56:24 MSK
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: context_info
    namespace: default
    user: minikube
  name: minikube
current-context: minikube
kind: Config
preferences: {}
users:
- name: minikube
  user:
    client-certificate: C:\Users\alese\.minikube\profiles\minikube\client.crt
    client-key: C:\Users\alese\.minikube\profiles\minikube\client.key
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl config current-context
minikube
```

Практическое задание

Сохранение вывода команды kubectl get nodes -o wide в файл nodes.txt

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl get nodes -o wide > nodes.txt
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ cat nodes.txt
NAME       STATUS   ROLES           AGE    VERSION   INTERNAL-IP    EXTERNAL-IP   OS-IMAGE                         KERNEL-VERSION
      CONTAINER-RUNTIME
minikube   Ready    control-plane   112m   v1.35.1   192.168.49.2   <none>        Debian GNU/Linux 12 (bookworm)   6.6.87.2-microsoft-standard-W
SL2   docker://29.2.1
```

2. Созданные ресурсы

2.1 Pods

Применение манифеста

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/01-pod-nginx.yaml
pod/goapp-pod created
```

Просмотр подов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
nginx-pod   1/1     Running   0          3m31s
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -o wide
NAME        READY   STATUS    RESTARTS   AGE     IP            NODE       NOMINATED NODE   READINESS GATES
nginx-pod   1/1     Running   0          3m33s   10.244.0.8    minikube   <none>           <none>

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods --show-labels
NAME        READY   STATUS    RESTARTS   AGE     LABELS
nginx-pod   1/1     Running   0          3m42s   app=nginx,lab=lab2
```

Детальная информация

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe pod nginx-pod
Name:             nginx-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Mon, 20 Apr 2026 17:59:12 +0300
Labels:           app=nginx
                  lab=lab2
Annotations:      <none>
Status:           Running
IP:               10.244.0.8
IPs:
  IP:  10.244.0.8
Containers:
  nginx:
    Container ID:   docker://cea9bf02ff2ecdd1d831f8433e3f961643b34bd40d0e017e4bdf3b448945c641
    Image:          nginx:alpine
    Image ID:       docker-pullable://nginx@sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Mon, 20 Apr 2026 17:59:27 +0300
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:        250m
      memory:     64Mi
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-m4mrq (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-m4mrq:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   Burstable
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  4m31s  default-scheduler  Successfully assigned default/nginx-pod to minikube
  Normal  Pulling    4m31s  kubelet            Pulling image "nginx:alpine"
  Normal  Pulled     4m16s  kubelet            Successfully pulled image "nginx:alpine" in 14.275s (14.275s including waiting). Image size: 6218
1394 bytes.
  Normal  Created    4m16s  kubelet            Container created
  Normal  Started    4m16s  kubelet            Container started
```

Логи

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs nginx-pod
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/20 14:59:27 [notice] 1#1: using the "epoll" event method
2026/04/20 14:59:27 [notice] 1#1: nginx/1.29.8
2026/04/20 14:59:27 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/04/20 14:59:27 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/04/20 14:59:27 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/20 14:59:27 [notice] 1#1: start worker processes
2026/04/20 14:59:27 [notice] 1#1: start worker process 30
2026/04/20 14:59:27 [notice] 1#1: start worker process 31
2026/04/20 14:59:27 [notice] 1#1: start worker process 32
2026/04/20 14:59:27 [notice] 1#1: start worker process 33
2026/04/20 14:59:27 [notice] 1#1: start worker process 34
2026/04/20 14:59:27 [notice] 1#1: start worker process 35
2026/04/20 14:59:27 [notice] 1#1: start worker process 36
2026/04/20 14:59:27 [notice] 1#1: start worker process 37
2026/04/20 14:59:27 [notice] 1#1: start worker process 38
2026/04/20 14:59:27 [notice] 1#1: start worker process 39
2026/04/20 14:59:27 [notice] 1#1: start worker process 40
2026/04/20 14:59:27 [notice] 1#1: start worker process 41
2026/04/20 14:59:27 [notice] 1#1: start worker process 42
2026/04/20 14:59:27 [notice] 1#1: start worker process 43
2026/04/20 14:59:27 [notice] 1#1: start worker process 44
2026/04/20 14:59:27 [notice] 1#1: start worker process 45
```

Выполнение команд в поде

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl exec -it nginx-pod -- sh
/ # curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit
```

Удаление пода

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl delete pod nginx-pod
pod "nginx-pod" deleted
```

Практическое задание

Манифест пода

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ cat 02-pod-go-app.yaml
apiVersion: v1
kind: Pod
metadata:
  name: go-app-pod
  labels:
    app: go
    lab: lab2
spec:
  imagePullSecrets:
    - name: ghcr-secret
  containers:
    - name: app
      image: ghcr.io/alelyak/containers-lab-1-app:v1.0.0
      ports:
        - containerPort: 8080
      resources:
        requests:
          memory: "64Mi"
          cpu: "250m"
        limits:
          memory: "128Mi"
          cpu: "500m"
```

Статус пода после запуска

```
Events:
  Type     Reason     Age                  From               Message
  ----     ------     ----                 ----               -------
  Normal   Scheduled  6m33s                default-scheduler  Successfully assigned default/go-app-pod to minikube
  Normal   Pulling    6m32s                kubelet            Pulling image "ghcr.io/alelyak/containers-lab-1-app:v1.0.0"
  Normal   Pulled     6m16s                kubelet            Successfully pulled image "ghcr.io/alelyak/containers-lab-1-app:v1.0.0" in 12.607s (1
6.445s including waiting). Image size: 17281029 bytes.
  Normal   Created    2m53s (x6 over 6m15s)  kubelet          Container created
  Normal   Started    2m53s (x6 over 6m14s)  kubelet          Container started
  Normal   Pulled     2m53s (x5 over 6m13s)  kubelet          Container image "ghcr.io/alelyak/containers-lab-1-app:v1.0.0" already present on mach
ine and can be accessed by the pod
  Warning  BackOff    81s (x7 over 6m11s)    kubelet          Back-off restarting failed container app in pod go-app-pod_default(e0b59a1d-0664-4231
-aec5-65bde550f1bf)
```

Сохранение логов в файл

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl logs go-app-pod > crash-logs.txt
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ cat crash-logs.txt
2026/04/20 16:14:21 Error creating table:dial tcp [::1]:5432: connect: connection refused
```

Работа с ReplicaSet

Создание ReplicaSet

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/02-replicaset-nginx.yaml
replicaset.apps/nginx-replicaset created
```

Просмотр ReplicaSet

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get rs
NAME               DESIRED   CURRENT   READY   AGE
nginx-replicaset   3         3         3       99s

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe rs nginx-replicaset
Name:         nginx-replicaset
Namespace:    default
Selector:     app=nginx
Labels:       app=nginx
              lab=lab2
Annotations:  <none>
Replicas:     3 current / 3 desired
Pods Status:  3 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=nginx
  Containers:
   nginx:
    Image:         nginx:alpine
    Port:          80/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Events:
  Type    Reason            Age   From                   Message
  ----    ------            ----  ----                   -------
  Normal  SuccessfulCreate  115s  replicaset-controller  Created pod: nginx-replicaset-2f89w
  Normal  SuccessfulCreate  115s  replicaset-controller  Created pod: nginx-replicaset-2vrzc
  Normal  SuccessfulCreate  115s  replicaset-controller  Created pod: nginx-replicaset-bzr8k
```

Просмотр созданных подов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=nginx
NAME                     READY   STATUS    RESTARTS   AGE
nginx-replicaset-2f89w   1/1     Running   0          3m40s
nginx-replicaset-2vrzc   1/1     Running   0          3m40s
nginx-replicaset-bzr8k   1/1     Running   0          3m40s
```

Масштабирование вручную

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl scale rs nginx-replicaset --replicas=5
replicaset.apps/nginx-replicaset scaled
```

Наблюдение вручную

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=nginx -w
NAME                     READY   STATUS    RESTARTS   AGE
nginx-replicaset-2f89w   1/1     Running   0          5m48s
nginx-replicaset-2fvl9   1/1     Running   0          43s
nginx-replicaset-2vrzc   1/1     Running   0          5m48s
nginx-replicaset-bzr8k   1/1     Running   0          5m48s
nginx-replicaset-j4wtm   1/1     Running   0          43s
```

Эксперимент с самовосстановлением

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl delete pod nginx-replicaset-2f89w
pod "nginx-replicaset-2f89w" deleted

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=nginx -w
NAME                     READY   STATUS    RESTARTS   AGE
nginx-replicaset-2fvl9   1/1     Running   0          3m8s
nginx-replicaset-2vrzc   1/1     Running   0          8m13s
nginx-replicaset-bzr8k   1/1     Running   0          8m13s
nginx-replicaset-cnswn   1/1     Running   0          20s
nginx-replicaset-j4wtm   1/1     Running   0          3m8s
```

2.2 Deployments

Применение манифеста

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/03-deployment-app.yaml
deployment.apps/go-app-deployment created

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр Deployment

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
go-app-deployment   0/2     2            0           107s
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe deployment go-app-deployment
Name:                   go-app-deployment
Namespace:              default
CreationTimestamp:      Mon, 20 Apr 2026 20:52:34 +0300
Labels:                 app=go-app
                        lab=lab2
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=go-app
Replicas:               2 desired | 2 updated | 2 total | 0 available | 2 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  0 max unavailable, 1 max surge
Pod Template:
  Labels:  app=go-app
  Containers:
   go-app:
    Image:      ghcr.io/alelyak/containers-lab-1-app:v1.0.0
    Port:       8080/TCP
    Host Port:  0/TCP
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:      250m
      memory:   64Mi
    Liveness:   http-get http://:8080/health delay=15s timeout=1s period=20s #success=1 #failure=3
    Readiness:  http-get http://:8080/health delay=5s timeout=1s period=10s #success=1 #failure=3
    Environment:
      DB_HOST:      postgres-service
      DB_PORT:      5432
      DB_USER:      postgres
      DB_PASSWORD:  secret
      DB_NAME:      myapp
    Mounts:         <none>
  Volumes:          <none>
  Node-Selectors:   <none>
  Tolerations:      <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      False   MinimumReplicasUnavailable
  Progressing    True    ReplicaSetUpdated
OldReplicaSets:  <none>
NewReplicaSet:   go-app-deployment-6c6c497df4 (2/2 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  2m27s  deployment-controller  Scaled up replica set go-app-deployment-6c6c497df4 from 0 to 2
```

Просмотр созданного ReplicaSet

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get rs -l app=go-app
NAME                         DESIRED   CURRENT   READY   AGE
go-app-deployment-6c6c497df4   2         2         0       3m29s
```

Просмотр подов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=go-app
NAME                               READY   STATUS    RESTARTS        AGE
go-app-deployment-6c6c497df4-8tg9t   0/1     Running   5 (95s ago)     3m42s
go-app-deployment-6c6c497df4-8z9lq   0/1     Error     5 (105s ago)    3m42s
```

Просмотр истории обновлений

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl rollout history deployment go-app-deployment
deployment.apps/go-app-deployment
REVISION  CHANGE-CAUSE
1         <none>
```

Масштабирование через Deployment

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl scale deployment go-app-deployment --replicas=3
deployment.apps/go-app-deployment scaled
```

Обновление образа

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl set image deployment/go-app-deployment go-app=ghcr.io/alelyak/containers-lab-1-app:v2.0.0
deployment.apps/go-app-deployment image updated
```

Откат изменений

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl rollout undo deployment go-app-deployment
deployment.apps/go-app-deployment rolled back
```

Практическое задание

Deployment для PostgreSQL

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1/k8s (lab2/kubernetes-basics)
$ cat 03-deployment-postgresql.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: postgres-deployment
  labels:
    app: postgres
    lab: lab2
spec:
  replicas: 1
  selector:
    matchLabels:
      app: postgres
  template:
    metadata:
      labels:
        app: postgres
    spec:
      containers:
      - name: postgres
        image: postgres:alpine
        ports:
        - containerPort: 5432
        env:
        - name: POSTGRES_USER
          value: "postgres"
        - name: POSTGRES_PASSWORD
          value: "secret"
        - name: POSTGRES_DB
          value: "myapp"
```

Deployment для PostgreSQL

Применение манифеста

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/03-deployment-postgresql.yaml
deployment.apps/postgres-deployment created

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
```

Статус Deployments

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
go-app-deployment     0/3     3            0           19m
postgres-deployment   1/1     1            1           5m45s
```

Статус подов и логи Go-App

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods
NAME                                  READY   STATUS             RESTARTS        AGE
go-app-deployment-5748d877dc-crczp    0/1     ImagePullBackOff   0               18m
go-app-deployment-6c6c497df4-8tg9t    0/1     CrashLoopBackOff   9 (95s ago)     24m
go-app-deployment-6c6c497df4-8z91q    0/1     CrashLoopBackOff   9 (102s ago)    24m
go-app-deployment-6c6c497df4-lrdsz    0/1     CrashLoopBackOff   8 (2m26s ago)   18m
go-app-pod                            0/1     CrashLoopBackOff   29 (103s ago)   129m
nginx-replicaset-2fvl9                1/1     Running            0               31m
nginx-replicaset-2vrzc                1/1     Running            0               36m
nginx-replicaset-bzr8k                1/1     Running            0               36m
nginx-replicaset-cnswn                1/1     Running            0               28m
nginx-replicaset-j4wtm                1/1     Running            0               31m
postgres-deployment-c6686877b-964dc   1/1     Running            0               10m

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs go-app-deployment-6c6c497df4-lrdsz
2026/04/20 18:14:31 Error creating table:dial tcp: lookup postgres-service on 10.96.0.10:53: server misbehaving
```

2.3 Services

Создание сервисов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/04-service-postgres.yaml
service/postgres-service created

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/05-service-app.yaml
service/go-app-service created

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/06-service-nginx.yaml
service/nginx-service created
```

Просмотр сервисов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get svc
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
go-app-service     ClusterIP   10.103.110.181   <none>        8080/TCP       9m36s
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP        144m
nginx-service      NodePort    10.105.130.10    <none>        80:30080/TCP   9m26s
postgres-service   ClusterIP   10.99.134.254    <none>        5432/TCP       9m43s
```

Детальная информация

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe svc postgres-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
Name:              postgres-service
Namespace:         default
Labels:            app=postgres
                   lab=lab2
Annotations:       <none>
Selector:          app=postgres
Type:              ClusterIP
IP Family Policy:  SingleStack
IP Families:       IPv4
IP:                10.99.134.254
IPs:               10.99.134.254
Port:              <unset>  5432/TCP
TargetPort:        5432/TCP
Endpoints:         10.244.0.15:5432
Session Affinity:  None
Events:            <none>
```

Проверка DNS-резолвинга внутри кластера

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl run test-pod --image=alpine:latest --rm -it --restart=Never -- sh
If you don't see a command prompt, try pressing enter.
/ # apk add curl postgresql-client
( 1/17) Installing brotli-libs (1.2.0-r0)
( 2/17) Installing c-ares (1.34.6-r0)
( 3/17) Installing libunistring (1.4.1-r0)
( 4/17) Installing libidn2 (2.3.8-r0)
( 5/17) Installing nghttp2-libs (1.68.0-r0)
( 6/17) Installing nghttp3 (1.13.1-r0)
( 7/17) Installing libpsl (0.21.5-r3)
( 8/17) Installing zstd-libs (1.5.7-r2)
( 9/17) Installing libcurl (8.17.0-r1)
(10/17) Installing curl (8.17.0-r1)
(11/17) Installing postgresql-common (1.2-r2)
  Executing postgresql-common-1.2-r2.pre-install
(12/17) Installing lz4-libs (1.10.0-r0)
(13/17) Installing libpq (18.3-r0)
(14/17) Installing ncurses-terminfo-base (6.5_p20251123-r0)
(15/17) Installing libncursesw (6.5_p20251123-r0)
(16/17) Installing readline (8.3.1-r0)
(17/17) Installing postgresql18-client (18.3-r0)
Executing busybox-1.37.0-r30.trigger
Executing postgresql-common-1.2-r2.trigger
* Setting postgresql18 as the default version
* find: /usr/share/man: No such file or directory
OK: 17.6 MiB in 33 packages
/ # curl go-app-service:8080/api/health
{"status":"healthy"}
/ # pg_isready -h postgres-service
postgres-service:5432 - accepting connections
/ # exit
pod "test-pod" deleted
```

Валидация YAML

Проверка синтаксиса YAML с помощью онлайн-сервиса yamllint

YAML Lint

Paste in your YAML and click "Go" - we'll tell you if it's valid or not, and give you a nice clean UTF-8 version of it.

```yaml
kind: Pod
metadata:
  name: go-app-pod
  labels:
    app: go
    lab: lab2
spec:
  imagePullSecrets:
    - name: ghcr-secret
  containers:
    - name: app
      image: ghcr.io/alelyak/containers-lab-1-app:v1.0.0
      ports:
        - containerPort: 8080
      resources:
        requests:
          memory: 64Mi
          cpu: 250m
        limits:
          memory: 128Mi
```

Go ☑ Reformat (strips comments) ☑ Resolve aliases

Valid YAML!

Сухая проверка (dry-run) в Kubernetes

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/03-deployment-app.yaml --dry-run=client
deployment.apps/go-app-deployment configured (dry run)
```

Получение объяснения по полям

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain pod
KIND:       Pod
VERSION:    v1

DESCRIPTION:
    Pod is a collection of containers that can run on a host. This resource is
    created by clients and scheduled onto hosts.

FIELDS:
  apiVersion    <string>
    APIVersion defines the versioned schema of this representation of an object.
    Servers should convert recognized schemas to the latest internal value, and
    may reject unrecognized values. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources

  kind  <string>
    Kind is a string value representing the REST resource this object
    represents. Servers may infer this from the endpoint the client submits
    requests to. Cannot be updated. In CamelCase. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds

  metadata      <ObjectMeta>
    Standard object's metadata. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata

  spec  <PodSpec>
    Specification of the desired behavior of the pod. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status

  status        <PodStatus>
    Most recently observed status of the pod. This data may not be up to date.
    Populated by the system. Read-only. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain deployment.spec
GROUP:      apps
KIND:       Deployment
VERSION:    v1

FIELD: spec <DeploymentSpec>


DESCRIPTION:
    Specification of the desired behavior of the Deployment.
    DeploymentSpec is the specification of the desired behavior of the
    Deployment.

FIELDS:
  minReadySeconds       <integer>
    Minimum number of seconds for which a newly created pod should be ready
    without any of its container crashing, for it to be considered available.
    Defaults to 0 (pod will be considered available as soon as it is ready)

  paused        <boolean>
    Indicates that the deployment is paused.

  progressDeadlineSeconds       <integer>
    The maximum time in seconds for a deployment to make progress before it is
    considered to be failed. The deployment controller will continue to process
    failed deployments and a condition with a ProgressDeadlineExceeded reason
    will be surfaced in the deployment status. Note that progress will not be
    estimated during the time a deployment is paused. Defaults to 600s.

  replicas      <integer>
    Number of desired pods. This is a pointer to distinguish between explicit
    zero and not specified. Defaults to 1.

  revisionHistoryLimit  <integer>
    The number of old ReplicaSets to retain to allow rollback. This is a pointer
    to distinguish between explicit zero and not specified. Defaults to 10.

  selector      <LabelSelector> -required-
    Label selector for pods. Existing ReplicaSets whose pods are selected by
    this will be the ones affected by this deployment. It must match the pod
    template's labels.

  strategy      <DeploymentStrategy>
    The deployment strategy to use to replace existing pods with new ones.

  template      <PodTemplateSpec> -required-
    Template describes the pods that will be created. The only allowed
    template.spec.restartPolicy value is "Always".
```

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain deployment.spec.template.spec.containers
GROUP:      apps
KIND:       Deployment
VERSION:    v1

FIELD: containers <[]Container>


DESCRIPTION:
    List of containers belonging to the pod. Containers cannot currently be
    added or removed. There must be at least one container in a Pod. Cannot be
    updated.
    A single application container that you want to run within a pod.

FIELDS:
  args  <[]string>
    Arguments to the entrypoint. The container image's CMD is used if this is
    not provided. Variable references $(VAR_NAME) are expanded using the
    container's environment. If a variable cannot be resolved, the reference in
    the input string will be unchanged. Double $$ are reduced to a single $,
    which allows for escaping the $(VAR_NAME) syntax: i.e. "$$(VAR_NAME)" will
    produce the string literal "$(VAR_NAME)". Escaped references will never be
    expanded, regardless of whether the variable exists or not. Cannot be
    updated. More info:
    https://kubernetes.io/docs/tasks/inject-data-application/define-command-argument-container/#running-a-command-in-a-shell

  command       <[]string>
    Entrypoint array. Not executed within a shell. The container image's
    ENTRYPOINT is used if this is not provided. Variable references $(VAR_NAME)
    are expanded using the container's environment. If a variable cannot be
    resolved, the reference in the input string will be unchanged. Double $$ are
    reduced to a single $, which allows for escaping the $(VAR_NAME) syntax:
    i.e. "$$(VAR_NAME)" will produce the string literal "$(VAR_NAME)". Escaped
    references will never be expanded, regardless of whether the variable exists
    or not. Cannot be updated. More info:
    https://kubernetes.io/docs/tasks/inject-data-application/define-command-argument-container/#running-a-command-in-a-shell
```

Валидация с помощью kubeconform (это усовершенствованная версия kubeval, который устарел)

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubeconform.exe -v
v0.7.0

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubeconform -verbose k8s/*.yaml
k8s/03-deployment-app.yaml - Deployment go-app-deployment is valid
k8s/03-deployment-postgresql.yaml - Deployment postgres-deployment is valid
k8s/04-service-postgres.yaml - Service postgres-service is valid
k8s/05-service-app.yaml - Service go-app-service is valid
k8s/06-service-nginx.yaml - Service nginx-service is valid
k8s/01-pod-nginx.yaml - Pod nginx-pod is valid
k8s/02-replicaset-nginx.yaml - ReplicaSet nginx-replicaset is valid
k8s/07-postgres-deployment.yaml - Deployment postgres-deployment is valid
k8s/02-pod-go-app.yaml - Pod go-app-pod is valid
k8s/08-nginx-deployment.yaml - Deployment nginx-deployment is valid
k8s/07-postgres-deployment.yaml - ConfigMap postgres-init-sql is valid
k8s/08-nginx-deployment.yaml - ConfigMap nginx-config is valid
k8s/08-nginx-deployment.yaml - ConfigMap nginx-html is valid
```

Отладка приложений

Просмотр событий в кластере

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get events --all-namespaces --sort-by='.lastTimestamp'
NAMESPACE     LAST SEEN   TYPE      REASON    OBJECT                               MESSAGE
kube-system   52m         Normal    BackOff   pod/metrics-server-9d74bb658-hqpfx   Back-off pulling image "regi
stry.k8s.io/metrics-server/metrics-server:v0.8.1@sha256:b2d2efaf5ac3b366ed0f839d2412a2c4279d4fc2a2a733f12c52133
faed36c41"
kube-system   52m         Warning   Failed    pod/metrics-server-9d74bb658-hqpfx   Error: ImagePullBackOff
```

Просмотр логов конкретного пода

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs -l app=go-app
2026/04/20 18:47:39 Server starting on port 8080
2026/04/20 18:47:39 Server starting on port 8080
```

Просмотр логов предыдущей инстанции пода

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs go-app-deployment-86cd6fbc4b-lttmk
2026/04/20 18:47:39 Server starting on port 8080
```

Порт-форвардинг для доступа к приложению без Service

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$kubectl port-forward pod/go-app-deployment-86cd6fbc4b-jgkt5 8080:8080
Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
```

Интерактивная отладка - запуск временного пода в той же сети

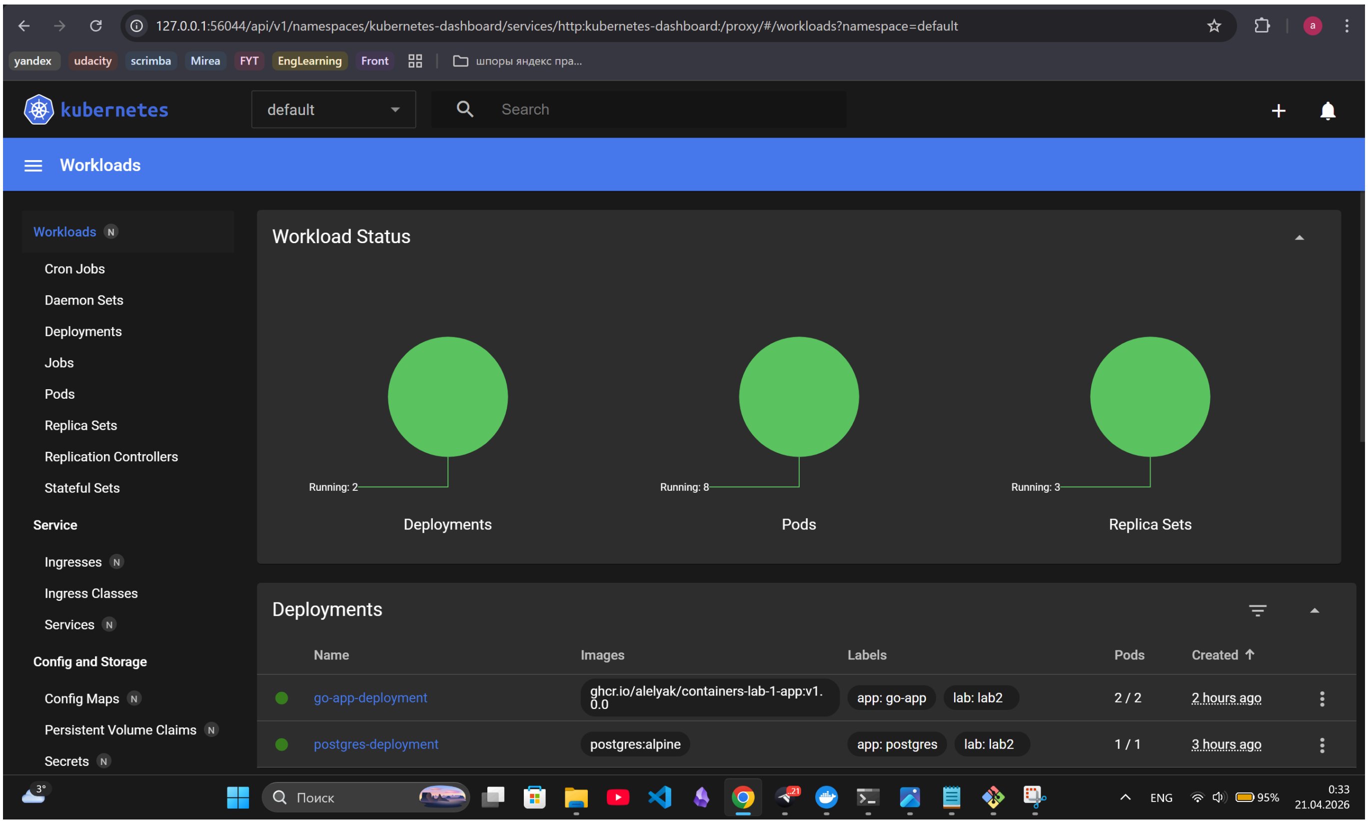
```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl run debug --image=nicolaka/netshoot:latest -it --rm -- bash
If you don't see a command prompt, try pressing enter.
debug:~# curl go-app-service:8080/health
404 page not found
debug:~# curl go-app-service:8080/api/health
{"status":"healthy"}
debug:~# nslookup postgres-service
;; Got recursion not available from 10.96.0.10
Server:         10.96.0.10
Address:        10.96.0.10#53

Name:   postgres-service.default.svc.cluster.local
Address: 10.99.134.254
;; Got recursion not available from 10.96.0.10

debug:~# |
```

Мониторинг через Dashboard

Запуск дашборда

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ minikube dashboard
* Enabling dashboard ...
  - Используется образ docker.io/kubernetesui/metrics-scraper:v1.0.8
  - Используется образ docker.io/kubernetesui/dashboard:v2.7.0
* Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

* Verifying dashboard health ...
* Launching proxy ...
* Verifying proxy health ...
* Opening http://127.0.0.1:56044/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
```

Открытый дашборд (1 упавший под - из практического задания в 2.1 Pods)



Последовательный запуск всех компонентов

Применение всех манифестов

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/
pod/nginx-pod created
pod/go-app-pod created
replicaset.apps/nginx-replicaset configured
deployment.apps/go-app-deployment unchanged
deployment.apps/postgres-deployment unchanged
service/postgres-service unchanged
service/go-app-service unchanged
service/nginx-service unchanged
deployment.apps/postgres-deployment configured
configmap/postgres-init-sql created
deployment.apps/nginx-deployment created
configmap/nginx-config created
configmap/nginx-html created
```

Проверка статуса

```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get all
NAME                                        READY   STATUS             RESTARTS      AGE
pod/go-app-deployment-86cd6fbc4b-jgkt5      1/1     Running            0             175m
pod/go-app-deployment-86cd6fbc4b-lttmk      1/1     Running            0             175m
pod/go-app-pod                              0/1     CrashLoopBackOff   4 (49s ago)   2m31s
pod/nginx-deployment-6f7bb479f4-2r26k       1/1     Running            0             2m28s
pod/nginx-deployment-6f7bb479f4-4jvdt       1/1     Running            0             2m22s
pod/postgres-deployment-75fc8ccd7-j2z8d     1/1     Running            0             2m29s

NAME                         TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/go-app-service       ClusterIP   10.103.110.181   <none>        8080/TCP       3h20m
service/kubernetes           ClusterIP   10.96.0.1        <none>        443/TCP        5h35m
service/nginx-service        NodePort    10.105.130.10    <none>        80:30080/TCP   3h20m
service/postgres-service     ClusterIP   10.99.134.254    <none>        5432/TCP       3h20m

NAME                                  READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/go-app-deployment     2/2     2            2           175m
deployment.apps/nginx-deployment      2/2     2            2           2m29s
deployment.apps/postgres-deployment   1/1     1            1           3h36m

NAME                                            DESIRED   CURRENT   READY   AGE
replicaset.apps/go-app-deployment-86cd6fbc4b    2         2         2       175m
replicaset.apps/nginx-deployment-6f7bb479f4     2         2         2       2m29s
replicaset.apps/nginx-replicaset                0         0         0       4h2m
replicaset.apps/postgres-deployment-75fc8ccd7   1         1         1       2m30s
replicaset.apps/postgres-deployment-c6686877b   0         0         0       3h36m
```

Ожидание готовности

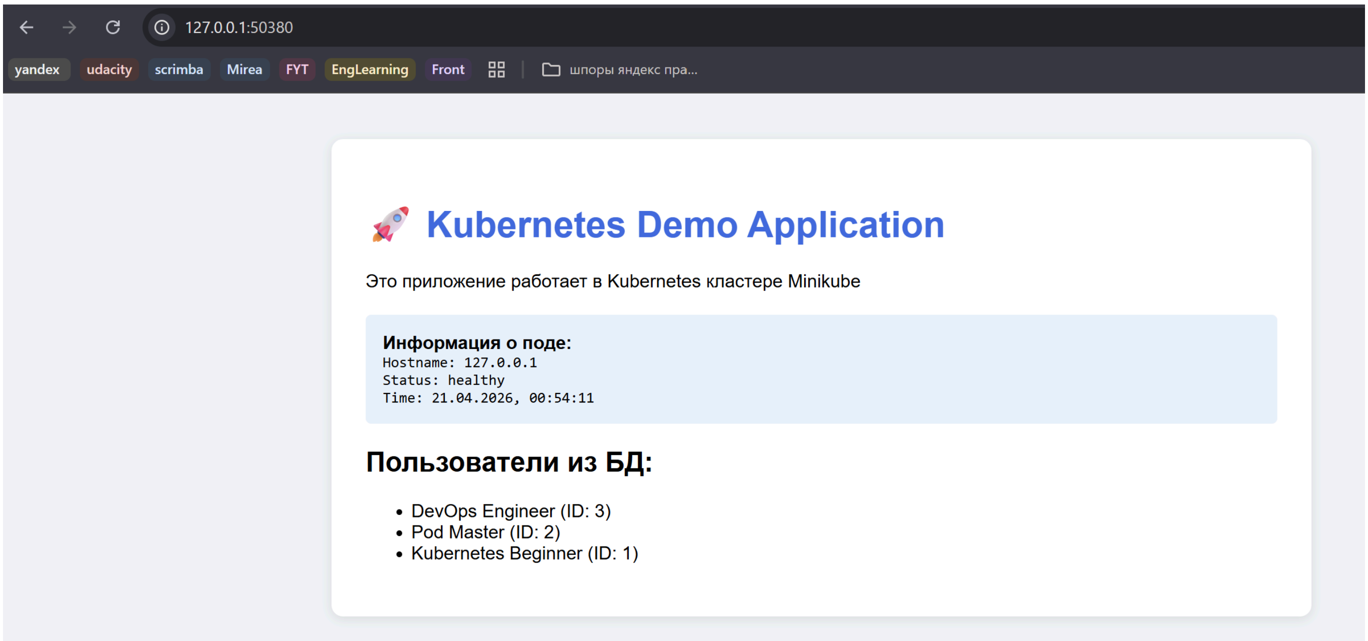
```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl wait --for=condition=ready pod -l app=postgres --timeout=60s
pod/postgres-deployment-75fc8ccd7-j2z8d condition met

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl wait --for=condition=ready pod -l app=go-app --timeout=60s
pod/go-app-deployment-86cd6fbc4b-jgkt5 condition met
pod/go-app-deployment-86cd6fbc4b-lttmk condition met

alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl wait --for=condition=ready pod -l app=nginx --timeout=60s
pod/nginx-deployment-6f7bb479f4-2r26k condition met
pod/nginx-deployment-6f7bb479f4-4jvdt condition met
```

Получение URL для доступа к приложению

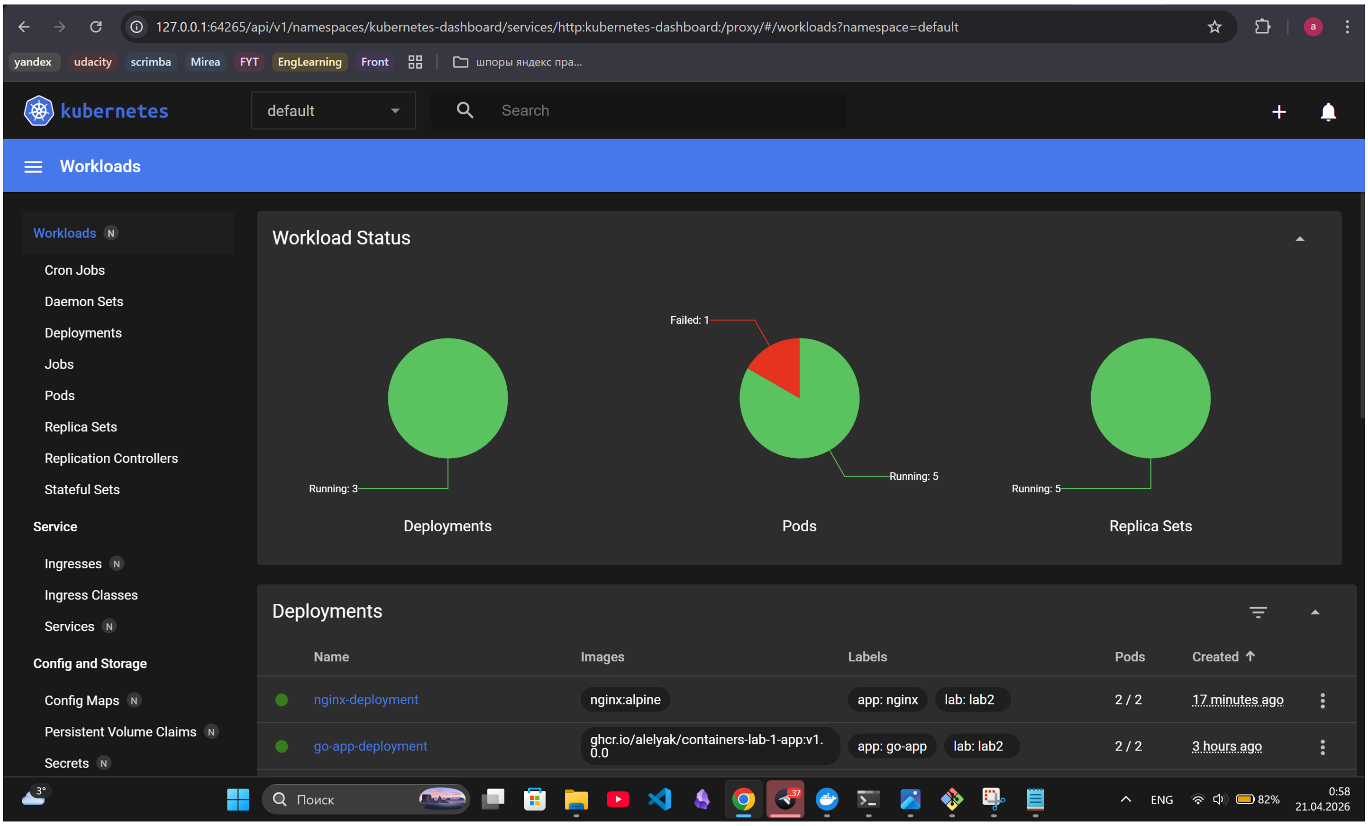
```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ minikube service nginx-service --url
http://127.0.0.1:57419
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

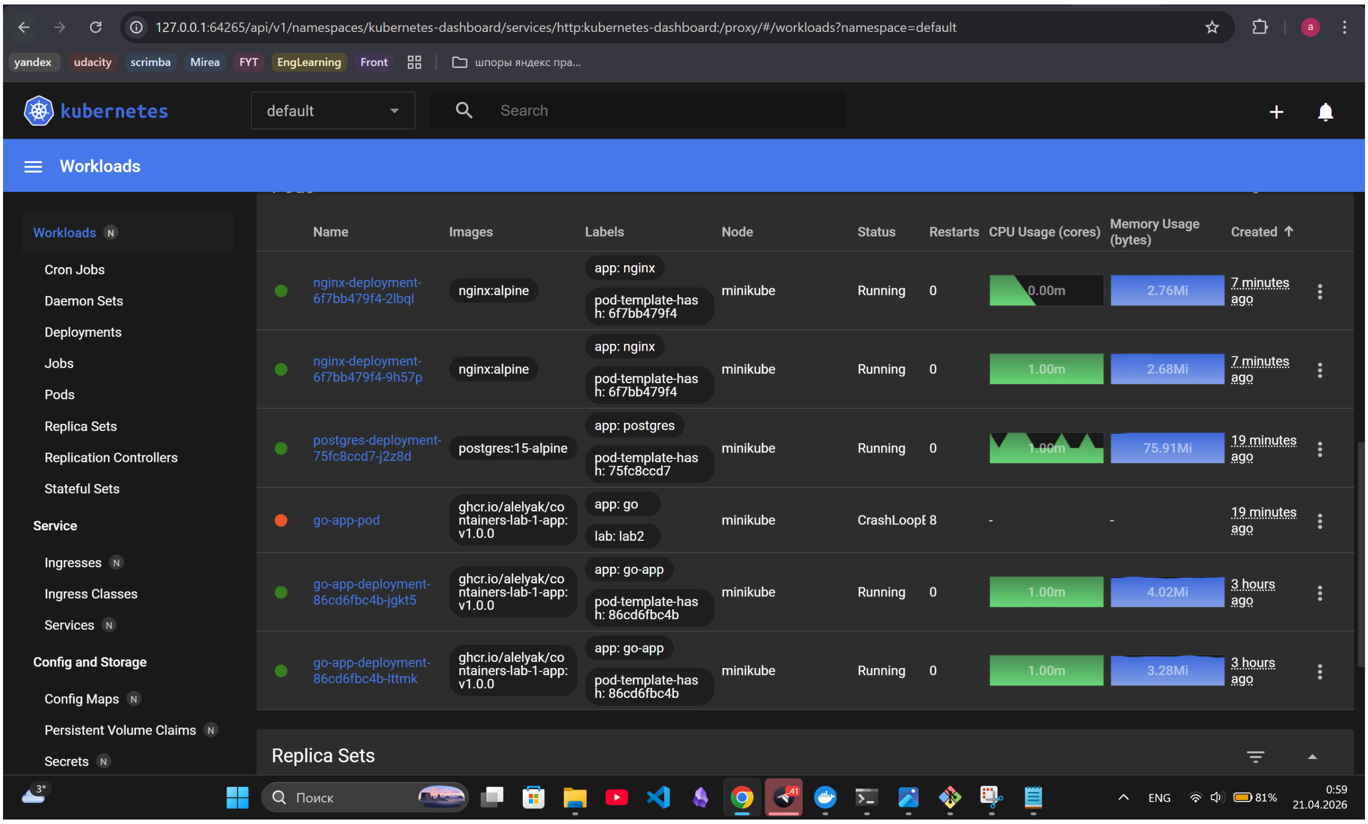
3. Скриншоты работы приложения

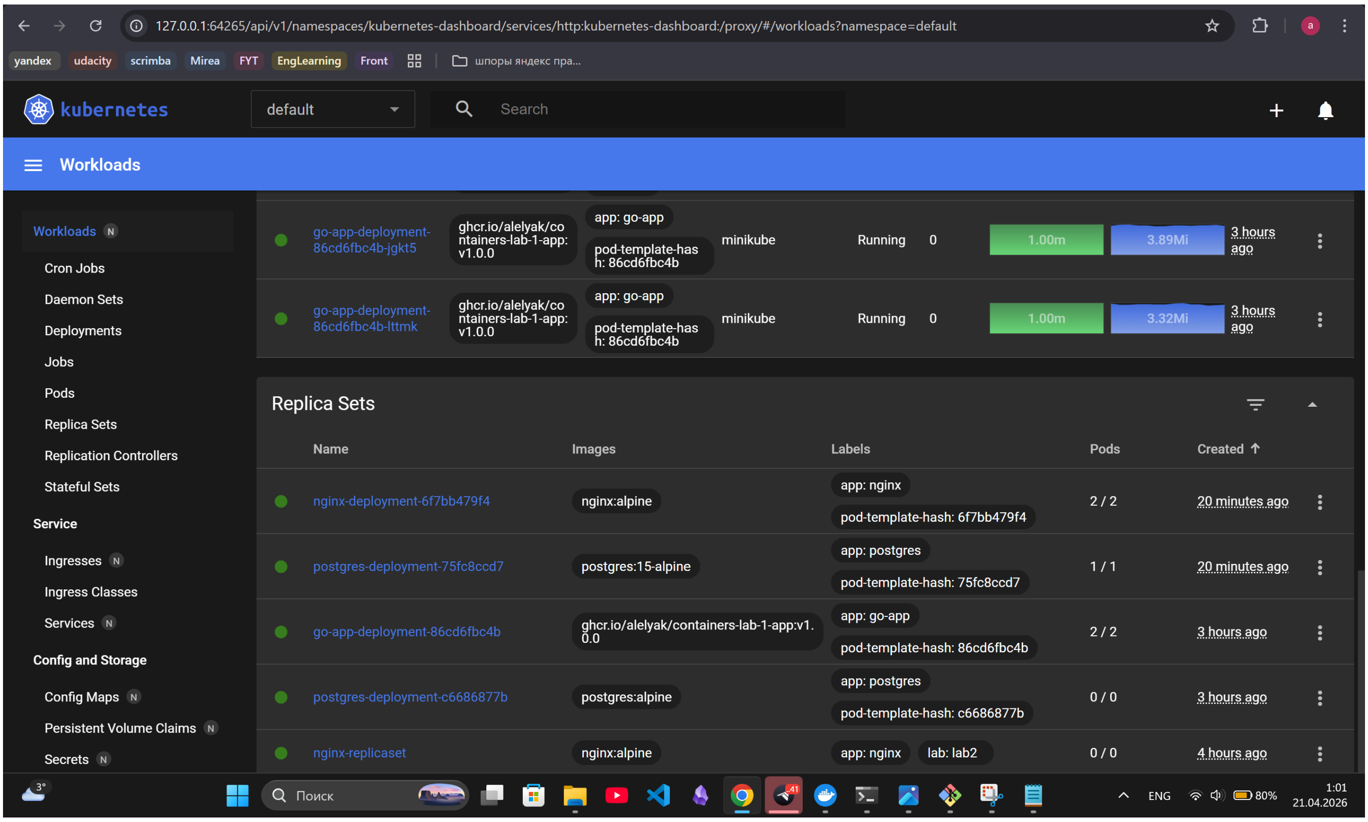
3.1 Главная страница



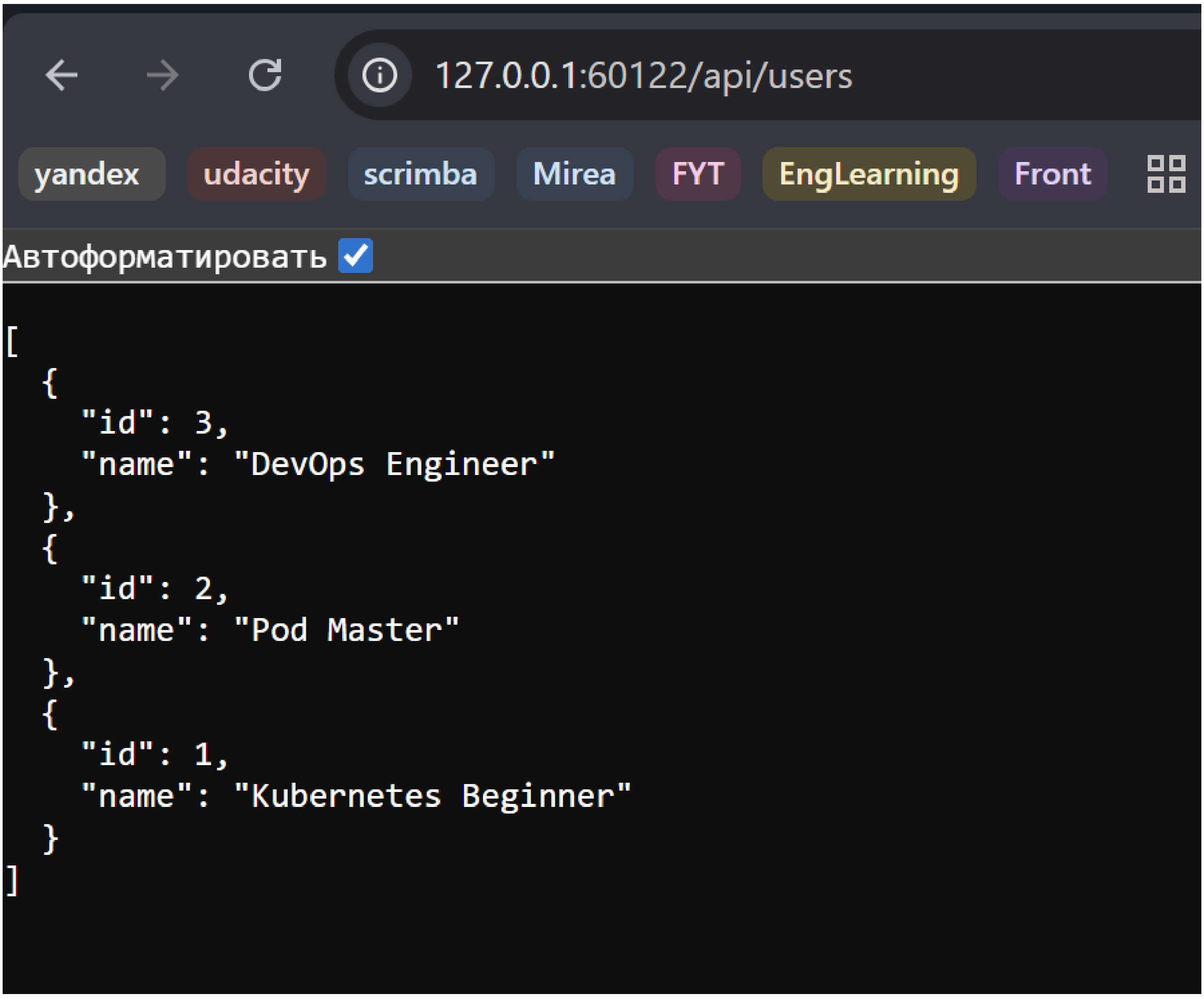
3.2 Дашборд Kubernetes


127.0.0.1:64265/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/#/workloads?namespace=default
kubernetes
default
Search
Workloads
Workloads N
Cron Jobs
Daemon Sets
Deployments
Jobs
Pods
Replica Sets
Replication Controllers
Stateful Sets
Service
Ingresses N
Ingress Classes
Services N
Config and Storage
Config Maps N
Persistent Volume Claims N
Secrets N
Workload Status
Running: 3
Deployments
Failed: 1
Running: 5
Pods
Running: 5
Replica Sets
Deployments
Name
Images
Labels
Pods
Created
nginx-deployment
nginx:alpine
app: nginx
lab: lab2
2 / 2
17 minutes ago
go-app-deployment
ghcr.io/alelyak/containers-lab-1-app:v1.0.0
app: go-app
lab: lab2
2 / 2
3 hours ago
0:58
21.04.2026


127.0.0.1:64265/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/#/workloads?namespace=default
kubernetes
default
Search
Workloads
Name
Images
Labels
Node
Status
Restarts
CPU Usage (cores)
Memory Usage (bytes)
Created
nginx-deployment-6f7bb479f4-2lbql
nginx:alpine
app: nginx
pod-template-hash: 6f7bb479f4
minikube
Running
0
0.00m
2.76Mi
7 minutes ago
nginx-deployment-6f7bb479f4-9h57p
nginx:alpine
app: nginx
pod-template-hash: 6f7bb479f4
minikube
Running
0
1.00m
2.68Mi
7 minutes ago
postgres-deployment-75fc8ccd7-j2z8d
postgres:15-alpine
app: postgres
pod-template-hash: 75fc8ccd7
minikube
Running
0
1.00m
75.91Mi
19 minutes ago
go-app-pod
ghcr.io/alelyak/containers-lab-1-app:v1.0.0
app: go
lab: lab2
minikube
CrashLoopE 8
19 minutes ago
go-app-deployment-86cd6fbc4b-jgkt5
ghcr.io/alelyak/containers-lab-1-app:v1.0.0
app: go-app
pod-template-hash: 86cd6fbc4b
minikube
Running
0
1.00m
4.02Mi
3 hours ago
go-app-deployment-86cd6fbc4b-lttmk
ghcr.io/alelyak/containers-lab-1-app:v1.0.0
app: go-app
pod-template-hash: 86cd6fbc4b
minikube
Running
0
1.00m
3.28Mi
3 hours ago
Replica Sets
0:59
21.04.2026



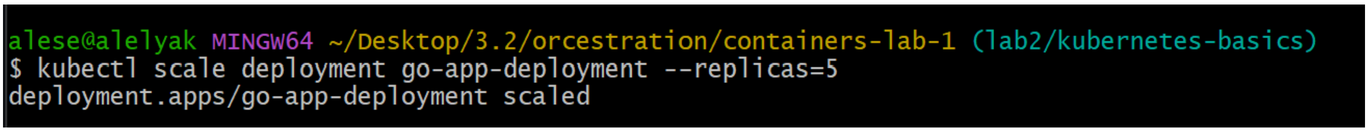
3.3 Результат GET /api/users



4. Эксперименты с масштабированием

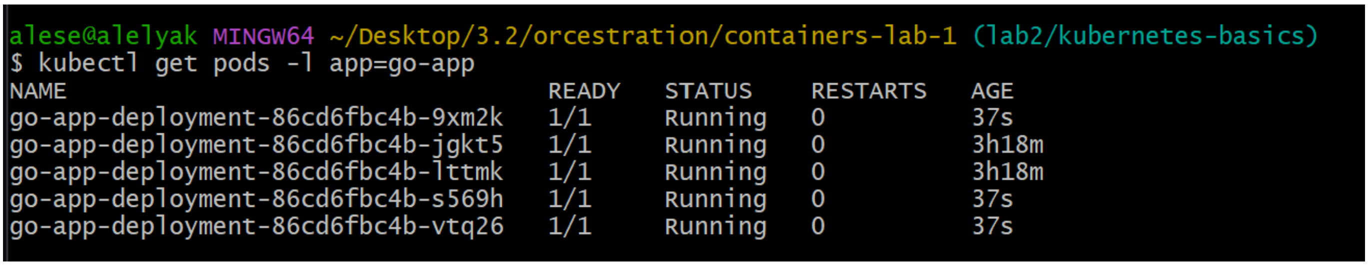
4.1 Масштабирование до 5 реплик

Масштабирование до 5 реплик



```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl scale deployment go-app-deployment --replicas=5
deployment.apps/go-app-deployment scaled
```

Статус подов Go-App после масштабирования



```
alese@alelyak MINGW64 ~/Desktop/3.2/orcestration/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=go-app
NAME                                 READY   STATUS    RESTARTS   AGE
go-app-deployment-86cd6fbc4b-9xm2k   1/1     Running   0          37s
go-app-deployment-86cd6fbc4b-jgkt5   1/1     Running   0          3h18m
go-app-deployment-86cd6fbc4b-lttmk   1/1     Running   0          3h18m
go-app-deployment-86cd6fbc4b-s569h   1/1     Running   0          37s
go-app-deployment-86cd6fbc4b-vtq26   1/1     Running   0          37s
```

4.2 Проверка распределения нагрузки

Команда PowerShell для нагрузочного тестирования

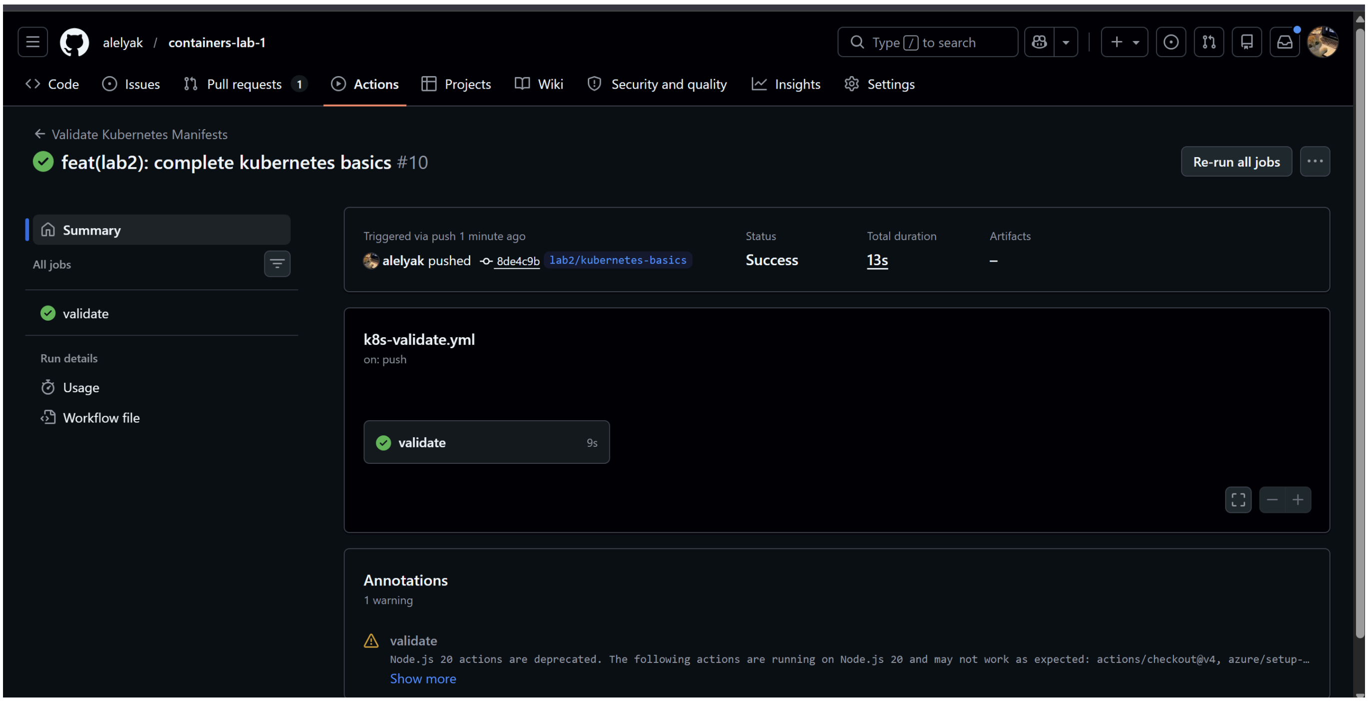
```
PS C:\Users\alese\Desktop\3.2\orcestration\containers-lab-1> for ($i=1; $i -le 100; $i++) {
>> curl http://localhost:49678/api/users
>> Start-Sleep -Milliseconds 100
>> }
```

Логи nginx с разных подов

```
PS C:\WINDOWS\system32> kubectl logs -l app=nginx --tail=10 -f
10.244.0.1 - - [20/Apr/2026:22:20:50 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:51 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:51 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:51 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:21:55:24 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:50380/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) C
hrome/147.0.0.0 Safari/537.36"
10.244.0.1 - - [20/Apr/2026:21:55:28 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:50380/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) C
hrome/147.0.0.0 Safari/537.36"
10.244.0.1 - - [20/Apr/2026:21:56:29 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:50380/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) C
hrome/147.0.0.0 Safari/537.36"
10.244.0.1 - - [20/Apr/2026:21:57:29 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:50380/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) C
hrome/147.0.0.0 Safari/537.36"
10.244.0.1 - - [20/Apr/2026:22:17:03 +0000] "GET /api/users HTTP/1.1" 200 103 "http://127.0.0.1:49678/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) C
hrome/147.0.0.0 Safari/537.36"
2026/04/20 22:17:03 [error] 30#30: *7 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 10.244.0.1, server: , request: "GET /favicon.ico HTTP/1.1",
 host: "127.0.0.1:49678", referrer: "http://127.0.0.1:49678/"
10.244.0.1 - - [20/Apr/2026:22:17:03 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://127.0.0.1:49678/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
 Chrome/147.0.0.0 Safari/537.36"
10.244.0.1 - - [20/Apr/2026:22:17:08 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:49678/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) C
hrome/147.0.0.0 Safari/537.36"
10.244.0.1 - - [20/Apr/2026:22:17:12 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari
/537.36"
10.244.0.1 - - [20/Apr/2026:22:18:25 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:52 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:52 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:52 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:53 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:53 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
10.244.0.1 - - [20/Apr/2026:22:20:53 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.8115"
```

5. GitHub Actions

5.1 Успешная валидация манифестов



6. Ответы на контрольные вопросы

1. В чем разница между Pod и Deployment?

Pod — минимальная единица развертывания в Kubernetes, Deployment — более высокоуровневый объект, который управляет подами, ReplicaSet обеспечивает стратегии обновления, отката и масштабирования.

2. Для чего нужен Service типа ClusterIP?

Для организации внутреннего взаимодействия между компонентами приложения, развёрнутого в одном кластере. Он создаёт внутренний IP-адрес внутри кластера, который позволяет другим подам внутри кластера взаимодействовать с сервисом, но недоступен из внешней сети.

3. Как ReplicaSet обеспечивает самовосстановление?

При удалении какого-то пода из ReplicaSet взамен него автоматически создаётся новый.

4. Что произойдет с приложением, если удалить под PostgreSQL?

Если удалить под PostgreSQL, то приложение продолжит работать, на странице в информации о поде в поле Status значение поменяется на unhealthy, пропадут записи о пользователях из БД. Затем через небольшой промежуток времени автоматически создадится новый под PostgreSQL, в поле Status снова будет записано healthy, и записи о пользователях снова появятся, они подтянутся из init-script.

7. Выводы

Из нового я узнала, как работать с Kubernetes и Minikube, валидировать YAML, проводить отладку приложений, проводить проверку DNS-резолвинга внутри кластера, мониторинг через Dashboard и как проводить нагрузочное тестирование.